EM Derivation

- not easy to optimize marginal likelihood

$\ell(\theta) = \log p(x|\theta)$ use LB.

q variational dist for z ($q(z)$ latent)

$$\ell(\theta) = \log p(x|\theta) = \sum_{z \in Z^n} \log p(x,z|\theta) = \log \sum_{z \in Z^n} q(z) \frac{p(x,z|\theta)}{q(z)}$$

Jensen's inequality ($\log$ = concave func)

$$= \log \mathbb{E}_{q(z)}\left[\frac{p(x,z|\theta)}{q(z)}\right] \geq \mathbb{E}_{q(z)}\left[\log \frac{p(x,z|\theta)}{q(z)}\right]$$

$$= \mathbb{E}_{q(z)}\left[\log p(x,z|\theta)\right] - \mathbb{E}_{q(z)}\left[\log q(z)\right]$$

$$= \mathbb{E}_{q(z)}\left[\log p(x,z|\theta)\right] - H(q)$$

Evidence Lower Bound $= ELBO(\theta, q)$

depends on unknown θ and latent $q(z)$

1) given $q(z)$, find θ that maximizes ELBO easier

$$Q(\theta) = \mathbb{E}_{q(z)}\left[\log p(x,z|\theta)\right] = \sum_{z \in Z^n} q(z) \log p(x,z|\theta)$$

entropy of q $H(q)$, max $Q(\theta)$ sufficient remark. ELBO

2) given fixed θ, finding $q(z)$ that max ELBO is necessary.

$$ELBO(\theta, q) = \mathbb{E}_{q(z)}\left[\log\left(\frac{p(x,z|\theta)}{q(z)}\right)\right]$$

$$= \mathbb{E}_{q(z)}\left[\log\left(\underbrace{\frac{p(x,z|\theta)}{p(z|x,\theta)}}\ \frac{p(z|x,\theta)}{q(z)}\right)\right]$$

$$= \mathbb{E}_{q(z)}\left[\log\left(p(x|\theta)\frac{p(z|x,\theta)}{q(z)}\right)\right]$$

$$= \mathbb{E}_{q(z)}\left[\log p(x|\theta)\right] - \mathbb{E}_{q(z)}\left[\log\left(\frac{q(z)}{p(z|x,\theta)}\right)\right]$$

$$= \log p(x|\theta) - KL\left(q \,\|\, p(z|x,\theta)\right)$$

$\therefore$ difference between marginal log likelihood & ELBO $= KL\left(q \,\|\, p(z|x,\theta)\right)$

$p(z|x,\theta)$ = posterior of latent variable (observed data, θ)

K.L nonnegative, only 0 when exactly the same, choice of $q(z)$ maximizes ELBO is precisely $p(z|x,\theta)$

ML optimization alternates between
- E-step / M-step
- 1) max. ELBO w.r.t θ, $q(z)$ fixed.
- 2) max. ELBO w.r.t. $q(z)$, θ fixed.

1) $\theta = \theta^{(0)}$

2) E-step: compute posterior of latent $p(z|x,\theta^{(t)})$ use to compute $Q(\theta)$ (expectation)

3) M-step: find new θ that max. $Q(\theta)$; $\theta^{(t+1)} = \arg\max_\theta Q(\theta)$ (maximize $Q(\theta)$)

where $q(z) = p(z|x,\theta^{(t)})$ maximizes ELBO at $\theta = \theta^{(t)}$ $\left(= \ell(\theta^{(t)})\right)$

Generalize: Θ collection of model parameters.

1) Derive E-step (find lower bound) use conditional posi. $p(z|x,\Theta^{(t)})$, and $Q(\Theta; \Theta^{(t)})$

$$Q(\Theta; \Theta^{(t)}) = \mathbb{E}_{p(z|x,\Theta^{(t)})}\left[\log p(x,z|\Theta)\right]$$

2) Derive M-step (update parameter) maximize LB)

find solution for Θ st. $\Theta^{(t+1)} = \arg\max_\Theta Q(\Theta; \Theta^{(t)})$ (update parameter)

ex) Failure rates of four labs.

x_i = observed failure time, $i = 1, \dots, n$ systems.

$z_i \in 1, \dots K$ latent, of system labs where i came from (unobserved)

use exponential where failure rate specific per batch k.

$x_i | z_i \overset{ind}{\sim} Exp(\lambda_k)$, $z_i \overset{iid}{\sim} Categorical(\pi)$ ← latent chosen

latent $\underline{\lambda} = (\lambda, \dots \lambda_K)$ and $\underline{\pi} = (\pi_1, \dots \pi_K)$ unknown. $\Theta = (\underline{\pi}, \underline{\lambda})$

E-step: $Q(\Theta; \Theta^{(t)}) = \mathbb{E}_z\left[\sum_i \log p(x_i, z_i|\Theta)\right]$ ← independent

$$= \sum_i \mathbb{E}_z\left[\log p(x_i|z_i, \underline{\lambda}) q(z_i|\underline{\pi})\right]$$

$$= \sum_i \left\{\underbrace{\mathbb{E}_z\left[\log p(x_i|z_i, \underline{\lambda})\right]}_{①} + \underbrace{\mathbb{E}_z\left[\log q(z_i|\underline{\pi})\right]}_{②}\right\}$$

① $$\sum_i \mathbb{E}_z\left[\log p(x_i|z_i, \underline{\lambda})\right] = \sum_i \sum_k p(z_i = k|x_i, \Theta^{(t)}) \log p(x_i|z_i = k, \lambda_k)$$

$$= \sum_i \sum_k \phi^t_{z_i}(k)\left(\log \lambda_k - \lambda_k x_i\right)$$

② $$\sum_i \mathbb{E}_z\left[\log q(z_i|\underline{\pi})\right] = \sum_i \sum_k p(z_i = k|x_i, \Theta^{(t)}) \log \pi_k$$

$$= \sum_i \sum_k \phi^t_{z_i}(k) \log \pi_k$$

$\phi^t_{z_i}(k) = p(z_i = k|x_i, \Theta)$

prob. of membership that ith obs x_i belongs to batch k.

independent of Θ (depends on $\Theta^{(t)}$)

$\sum_k \phi^t_{z_i}(k) = 1$ conditional

Bayes Rule: $$\phi^t_{z_i}(k) = p(z_i = k|\Theta^{(t)}) = \frac{p(x_i|z_i = k, \lambda^t_k)\, p(z_i = k|\pi^t_k)}{\sum_k p(x_i|z_i = k, \lambda^t_k)\, p(z_i = k|\pi^t_k)} = \frac{\lambda^t_k \exp\{-\lambda^t_k x_i\} \pi^t_k}{\sum_k \lambda^t_k \exp\{-\lambda^t_k x_i\} \pi^t_k}$$

$$\therefore Q(\Theta; \Theta^{(t)}) = \sum_i \sum_k \phi^t_{z_i}(k)\left(\log \lambda_k - \lambda_k x_i\right) + \sum_i \sum_k \phi^t_{z_i}(k) \log \pi_k$$

(functions of $\underline{\lambda}, \underline{\pi}, \underline{\lambda}^{(t)}, \underline{\pi}^{(t)}$)

M-step: take partial deriv. wrt each π_k parameter adding Lagrange mult. for α to constrain $\sum_k \pi_k = 1$

$$\frac{\partial}{\partial \pi_k}\left[Q^t + \alpha\left(1 - \sum_k \pi_k\right)\right] = \sum_i \frac{\phi^t_{z_i}(k)}{\pi_k} - \alpha \overset{\Delta}{=} 0 \Rightarrow \pi_k = \frac{\sum_i \phi^t_{z_i}(k)}{\alpha}$$

$$\because \sum_k \pi_k = 1,\ \phi^t_{z_i}(k) = 1 \Rightarrow \frac{\sum_k \sum_i \phi^t_{z_i}(k)}{\alpha} = 1 \Rightarrow \alpha = \sum_k \sum_i \phi^t_{z_i}(k) = \sum_i 1 = n$$

parameter update for π_k is: $$\pi_k = \frac{\sum_i \phi^t_{z_i}(k)}{n}$$

take partial for λ_k

$$\frac{\partial Q^t}{\partial \pi_k} = \sum_i \phi^t_{z_i}(k)\left(\lambda_k^{-1} - x_i\right) \overset{\Delta}{=} 0 \Rightarrow \lambda_k = \frac{\sum_i \phi^t_{z_i}(k)}{\sum_i \phi^t_{z_i}(k) x_i}$$

All together, two iterative update rules.

1) initialize $\pi_k^{(0)} = 1/K$, $\lambda_k^{(0)} = 1$

2) E-step: compute $\phi^t_{z_i}(k)$ and $Q^t(\Theta)$

$$\phi^t_{z_i}(k) = \frac{\lambda^t_k \exp\{-\lambda^t_k x_i\} \pi^t_k}{\sum_k \lambda^t_k \exp\{-\lambda^t_k x_i\} \pi^t_k}$$

3) M-step: update parameters.

$$\lambda^{(t+1)}_k = \frac{\sum_i \phi^t_{z_i}(k)}{\sum_i \phi^t_{z_i}(k) x_i}, \quad \pi^{(t+1)}_k = \frac{\sum_i \phi^t_{z_i}(k)}{n}$$